

1. Administrative & Study Identification

Field | Data Entry / Details ---|--- Full Study Title | A Phase Ib, multicenter, open-label platform study of select drug combinations in adult patients with lower risk (very low, low, or intermediate risk) myelodysplastic syndrome Short Title/Acronym | Phase Ib Study of Select Drug Combinations in Patients With Lower Risk MDS Protocol Number | CMBG453E12101 NCT Number | NCT04810611 Sponsor Name | Novartis Pharmaceuticals Investigational Product | MBG453 (sabatolimab), NIS793, canakinumab (ACZ885) Phase of Development | Phase Ib (Phase 1) Medical Monitor | Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To establish the recommended dose (RD), characterize general safety, and confirm the dose tolerability of select single agents (MBG453, NIS793, and canakinumab) and their combinations in patients with lower-risk Myelodysplastic Syndromes (MDS). Secondary Objectives: To evaluate preliminary efficacy (including reductions in transfusions, transfusion independence, and hemoglobin changes), pharmacokinetics, and pharmacodynamics of the study drugs.

2.2 Background & Rationale To address the unmet medical need in patients with lower-risk (very low, low, or intermediate risk) MDS who are refractory to, intolerant of, or ineligible/unable to receive standard-of-care (SOC) therapeutic options like lenalidomide and erythropoiesis-stimulating agents (ESAs). This platform study investigates whether novel immune-directed checkpoint modulators targeting TIM-3, TGF- β , and IL- β can safely provide clinical benefit and improve symptomatic cytopenias in this patient population.

3. Study Design & Methodology

3.1 Design Framework * Study Type: Interventional * Control Type: Single-arm (Multiple parallel arms evaluated independently) * Blinding: Open-label * Randomization: N/A (Adaptive platform study with dose confirmation/escalation parts and planned dose expansion)

3.2 Planned Enrollment & Duration * Targeted Enrollment: 33 patients * Number of Sites: Multicenter * Status: Terminated (Study was stopped early and did not progress into the expansion)

phase) * **Estimated Study Start:** March 2021 * **Estimated Primary Completion:** Stopped/Terminated early

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Signed informed consent must be obtained prior to participation. 2. Age ≥ 18 years at the time of signing the informed consent form. 3. Diagnosis of IPSS-R very low, low, or intermediate risk MDS with $\leq 10\%$ bone marrow blasts and one or more of the following: * Symptomatic anemia with hemoglobin < 10 g/dL that is relapsed/refractory to ESAs, or ESA-naïve with EPO ≥ 500 U/L. * Thrombocytopenia with platelets $< 30,000/\mu\text{L}$ or with clinically significant bleeding or bruising and platelets $< 50,000/\mu\text{L}$. * Neutropenia with an absolute neutrophil count (ANC) $< 500/\mu\text{L}$ or with recurrent and/or severe infections and an ANC that is $< 1000/\mu\text{L}$ (amenable to response assessments by IWG criteria). 4. Refractory to, intolerant of, or ineligible/unable to receive SOC therapeutic options including lenalidomide. 5. Eastern Cooperative Oncology Group (ECOG) performance status ≤ 2 . 6. Candidate for serial bone marrow aspirate and/or biopsy according to institution guidelines and willing to undergo procedures at screening, during, and at the end of therapy.

4.2 Exclusion Criteria 1. Systemic antineoplastic therapy (including cytotoxic chemotherapy, alpha-interferon, kinase inhibitors, etc.) or any experimental therapy within 14 days or 5 half-lives prior to the first dose of study treatment. 2. History of hypersensitivity to any of the study treatments or excipients or to drugs of similar chemical classes. 3. Diagnosis of chronic myelomonocytic leukemia (CMML) or myelodysplastic/myeloproliferative neoplasms (MDS/MPN). 4. Use of hematopoietic colony-stimulating growth factors (e.g., G-CSF, GM-CSF), thrombopoietin mimetics, or ESAs ≤ 2 weeks (or 5 half-lives) prior to the start of study treatment. 5. Systemic chronic corticosteroid therapy (>10 mg/day prednisone or equivalent) or any immunosuppressive therapy within 7 days of the first dose. (Topical, inhaled, nasal, and ophthalmic corticosteroids are permitted). 6. For arms containing canakinumab: Patients with ANC $< 500/\mu\text{L}$.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency:** Arm 1: MBG453 single agent; Arm 2: NIS793 single agent; Arm 3: canakinumab single agent; Arm 4: MBG453 + NIS793 combination (no patients treated); Arm 5: MBG453 + canakinumab combination. * **Route of Administration:** Intravenous (IV) / Subcutaneous (SC). * **Duration of Treatment:** Until disease progression, unacceptable toxicity, or study termination.

5.2 Concomitant & Prohibited Therapy * **Permitted:** Topical, inhaled, nasal, and ophthalmic corticosteroids. * **Prohibited:** Systemic chronic corticosteroid therapy (>10 mg/day prednisone) within 7 days of the first dose, other systemic antineoplastic therapies, ESAs, and strong immunosuppressive therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Bone Marrow Aspirate/Biopsy, Safety Labs
Baseline	Day 0 Enrollment, First Dose of Investigational Product	Interim	Weeks 3/4 \$\\pm\$ [Y] Safety Labs, Dose Escalation Review, Efficacy Assessment, Bone Marrow
End of Study	Disease Progression Final Safety Assessment, Bone Marrow Aspirate/Biopsy, Follow-up, IP Accountability		

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints * **Incidence of DLTs:** Incidence of dose-limiting toxicities (DLTs) during the first 2 cycles of treatment during the dose escalation/confirmation part. * **Dose Interruption Reduction:** Assessed as a measure of dose tolerability. * **Dose Intensity:** Assessed as a measure of dose tolerability.

7.2 Secondary & Exploratory Endpoints * Reduction in red blood cell (RBC) / platelet transfusions from baseline in transfusion-dependent patients. * Duration of transfusion independence lasting for \$\\ge\$ 8\$ weeks, \$\\ge\$ 12\$ weeks, \$\\ge\$ 16\$ weeks, and \$\\ge\$ 24\$ weeks in transfusion-dependent patients. * Change from baseline in hemoglobin (Hb) in transfusion-dependent and transfusion-independent patients. * Overall Response Rate (ORR) and Hematologic Improvement (HI) per IWG criteria. * Pharmacokinetics (Cmax, AUC) and Pharmacodynamics profile of the investigational agents.

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Assessed continuously via investigator evaluation, patient interviews, and clinical labs.
 - **Serious Adverse Events (SAEs):** Standard reporting timelines within 24 hours to the sponsor and regulatory authorities.
 - **Data Monitoring Committee (DMC):** Internal safety review committee governing the dose escalation/confirmation transitions.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Safety Set (all patients who received at least one dose of study treatment).
 - **Sample Size Justification:** Adaptive platform design targeted at 33 enrolled patients; expansion phases were originally planned but the study was terminated during the dose confirmation part.
 - **Handling of Missing Data:** Descriptive statistics without imputation for primary safety and tolerability analyses.
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** Conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** Periodic site monitoring and remote data verification.
 - **Audit Trail:** Standard electronic case report forms (eCRF) with robust audit trails and data clarification forms.
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11. Study Identifiers & Governance

- **Posting ID:** 100479642675566773
 - **Primary ID:** CMBG453E12101
 - **Registry Number:** NCT04810611
 - **Sponsor/Company:** Novartis / Novartis Pharmaceuticals
 - **Study Title:** Phase Ib Study of Select Drug Combinations in Patients With Lower Risk MDS
 - **Official Regulatory Title:** A Phase Ib, multicenter, open-label platform study of select drug combinations in adult patients with lower risk (very low, low, or intermediate risk) myelodysplastic syndrome
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12. Clinical Framework (MDS Specific)

- **Disease Area / Condition:** Myelodysplastic Syndromes (Lower Risk Myelodysplastic Syndrome)
 - **Diagnosis Threshold:** IPSS-R very low, low, or intermediate risk MDS with $\leq 10\%$ bone marrow blasts
 - **Medical Methodology:** Immune Checkpoint Modulation / Targeted Therapy
 - **Optimization Target:** Symptomatic anemia improvement, transfusion reduction, and disease control
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13. Study Procedures & Methodology

- **Management Pathway:** Targeted investigational therapy for lower-risk MDS

- **Education Protocol (HCP):** Site Initiation Visits and protocol training for novel platform combinations
 - **Education Protocol (Patient):** Informed consent process and study procedures education
 - **Quality Audit Mechanism:** Clinical monitoring and source data verification
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14. Observation & Follow-Up Schedule

- **Total Duration:** Until disease progression or study termination
 - **Onsite Visit Cadence:** Aligned with dosing cycles
 - **Remote Monitoring Frequency:** As per protocol monitoring plan
 - **Checklist Review Interval:** Ongoing safety and AE review prior to each dose
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15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	____	[DD-MMM-YYYY]				